BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Characters and Data Units

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: convert hexadecimal 3C to denary.

2. What is denary 42 in hex? 2B / 29 / A2 / 2A

3. Convert binary 00101010 to hex.

4. Convert hex 2A to eight-bit binary.

Today's objectives

Compare ASCII and Unicode; explain character sets and bits-to-bytes conversion.

Key vocabulary - Read, cover, recall

character set Maps characters to numerical codes	**ASCII** A smaller character repertoire
Unicode Supports many writing systems	**byte** Eight bits

Write one definition in your own words:

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Learn and practise

New learning - From character to binary

A character set maps a written character to a code, represented in bits.

Character	ASCII code	8-bit binary
A	65	01000001

B is code 66 (01000010); codes follow an ordered sequence.

ASCII versus Unicode

ASCII	Unicode
A smaller character repertoire	Supports many languages and scripts

Unicode encodings may use different numbers of bytes per character.

I Do - Worked calculation

12 characters use 8 bits each. Find the raw data size.

12 x 8 = 96 bits; 96 / 8 = 12 bytes.

We Do - Guided practice

18 characters at 8 bits each: work out the bits and bytes.

Bits = 18 x 8 = _____

Bytes = _____ / 8 = _____

Why is Unicode useful for a multilingual website?

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Think and write

Code First - Predict > Write > Test > Improve

Start on paper. Do not look at model answers until page 4.

```
characterCode = 65
print(characterCode)
```

Predict the output of the starter code.

Write both lines after changing 65 to 66.

Explain why the code displays 66, not the letter B.

You Do - Independent work

State how many bits are in a byte and a nibble. [2]

Explain an advantage of Unicode over ASCII. [2]

80 characters use 16 bits each. Calculate bytes. [3]

Thinking: Why is 'each Unicode character always uses one byte' wrong?

Exit Ticket - True or False?

Answer before you turn to the green-pen page.

1. A byte contains eight bits. ☐ T ☐ F
2. Unicode supports fewer characters than ASCII. ☐ T ☐ F
3. Every Unicode character uses exactly one byte. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 3 x 16 + 12 = 60 ☐

Q2 2A ☐

Q3 0010 | 1010 = 2A ☐

Q4 00101010 ☐

We Do - Guided practice

18 x 8 = 144 bits; / 8 = 18 bytes

Key vocabulary

character set ☐
Maps characters to numerical codes

ASCII ☐
A smaller character repertoire

Unicode ☐
Supports many writing systems

byte ☐
Eight bits

Code First - Answers

Predict ☐
65

Write ☐
characterCode = 66; print(characterCode)

Improve ☐
It prints a numeric value, not a character.

Exit Ticket - Explanations

1. True

2. False: Unicode supports more

3. False: encodings vary

You Do - Self-mark

Q1	Byte = 8 bits; nibble = 4 bits	☐
Q2	Unicode supports more languages and symbols	☐
Q3	80 x 16 = 1280 bits / 8 = 160 bytes	☐
Thinking	Encoding determines the number of bytes.	☐

Try the question again

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: